Manifolds



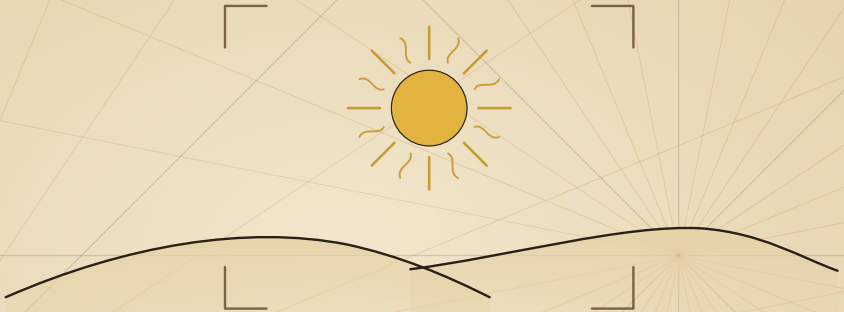
WEEKEND 4, FRIDAY MORNING

Carlos Cotrini

Friday 23 October 2026



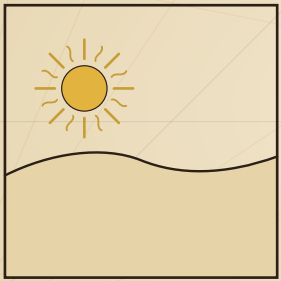
Every morning, the keeper watches the sun rise out of the sea.





Inland, someone holds up a phone and keeps the moment: a sunrise, saved as millions of pixels.

✳ *Predicting from a photo*



REGRESSION

Time of the day

CLASSIFICATION

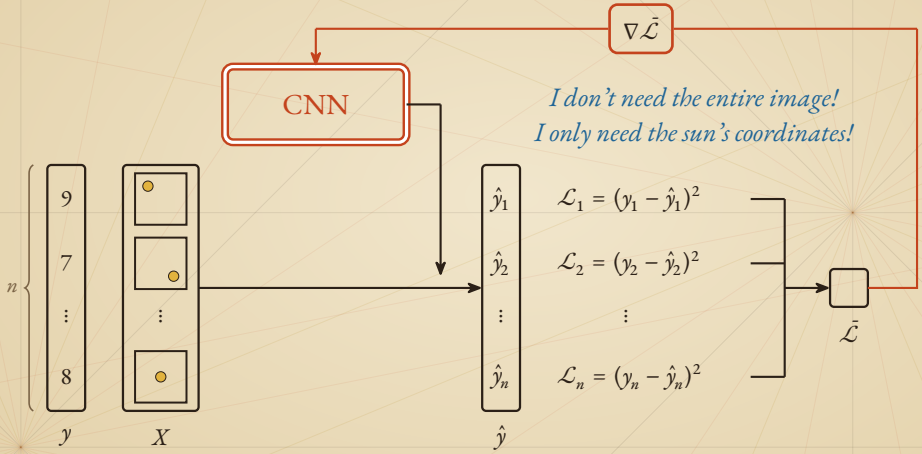
Season



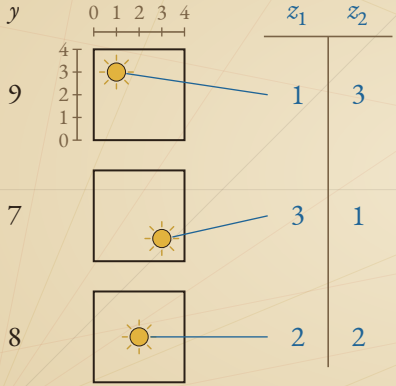
SECTION I

From pixels to coordinates

✳ *Vapnik's statistical learning*



✳ *The sun's coordinates*



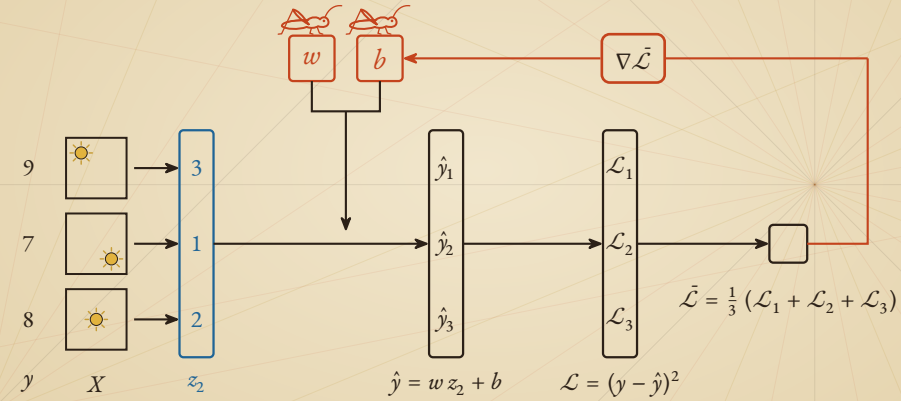
$$\hat{y} = w\, z_2 + b$$

$$\hat{y}_1 = w \cdot 3 + b$$

$$\hat{y}_2 = w \cdot 1 + b$$

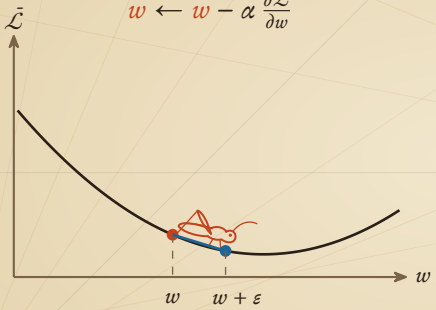
$$\hat{y}_3 = w \cdot 2 + b$$

✳ *Learning w and b*



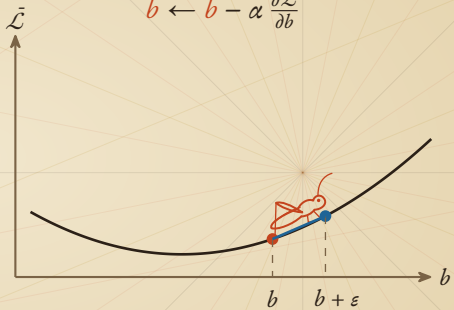
✳ *Each grasshopper measures its slope*

$$w \leftarrow w - \alpha \frac{\partial \bar{\mathcal{L}}}{\partial w}$$



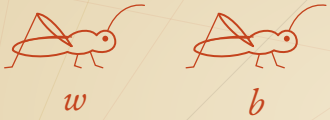
$$\frac{\partial \bar{\mathcal{L}}}{\partial w} \approx \frac{\bar{\mathcal{L}}(w + \varepsilon,\, b) - \bar{\mathcal{L}}(w,\, b)}{\varepsilon}$$

$$b \leftarrow b - \alpha \frac{\partial \bar{\mathcal{L}}}{\partial b}$$



$$\frac{\partial \bar{\mathcal{L}}}{\partial b} \approx \frac{\bar{\mathcal{L}}(w,\, b + \varepsilon) - \bar{\mathcal{L}}(w,\, b)}{\varepsilon}$$

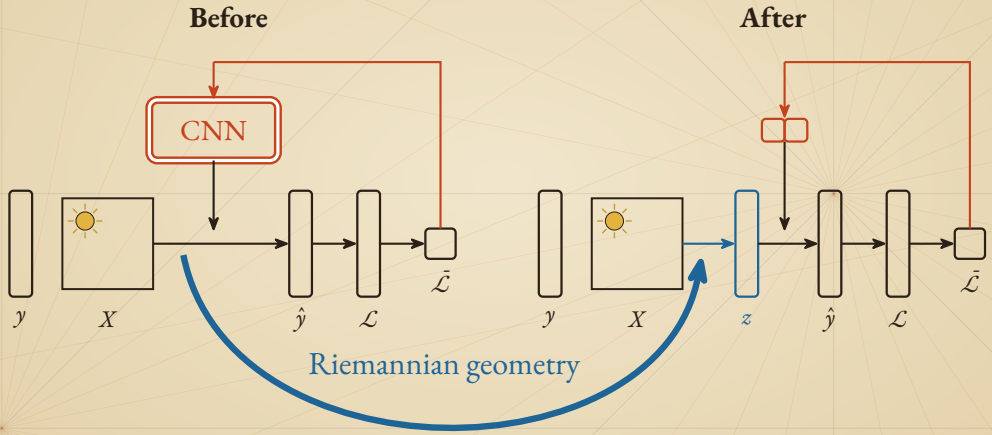
✳ *The grasshoppers, click by click*



Type where the two grasshoppers start,
then follow them, one click at a time.

https://eth-bmai-hs26.github.io/BMAI-PAGE/viz/we4/grasshopper-regression/

✳ *Before and after*





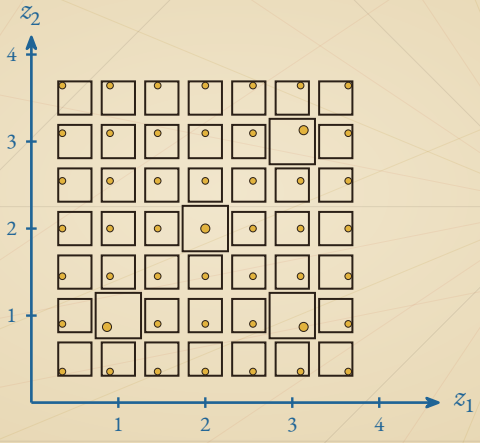
SECTION 2

Charts

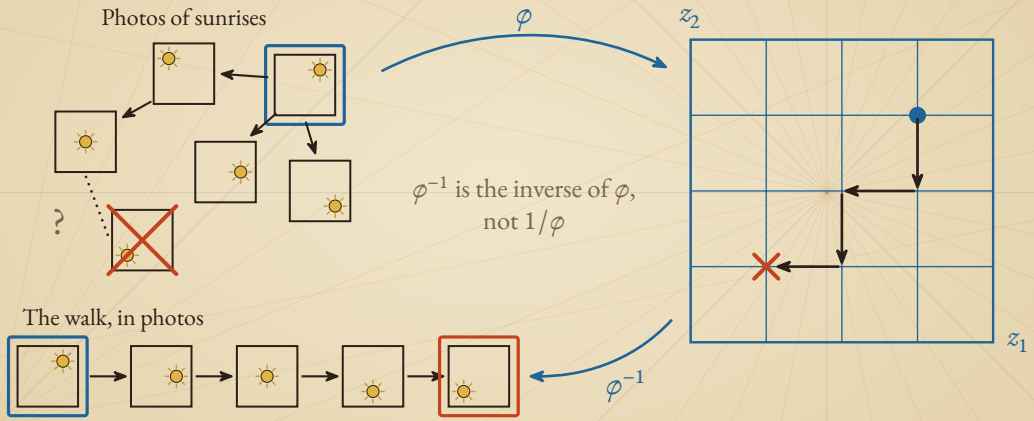
From the photo to its coordinates, and back



- *The photos, placed at their coordinates*



✳ *Walking through the sunrises*



✳ *Manifolds*



The set of all sunrises forms a *manifold*.

A *manifold* is any set of objects that admits a *homeomorphism* to a *navigation chart*.*

* *A white lie: in general, a manifold only needs such a chart around each of its points.*

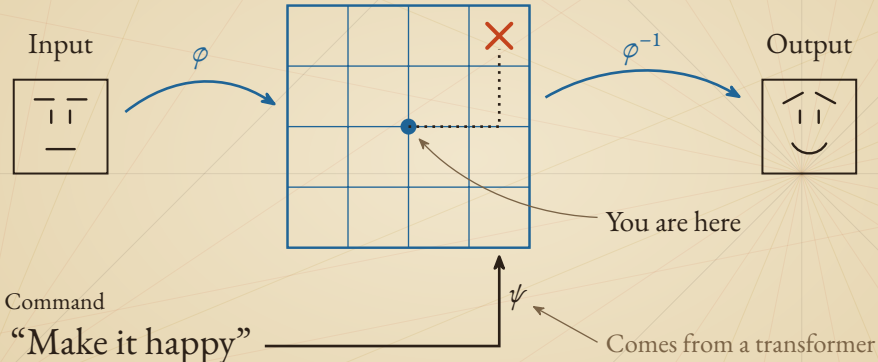
✳ *Another example*



✳ *A chart of faces*



✳ *The main application of Riemannian geometry to AI*

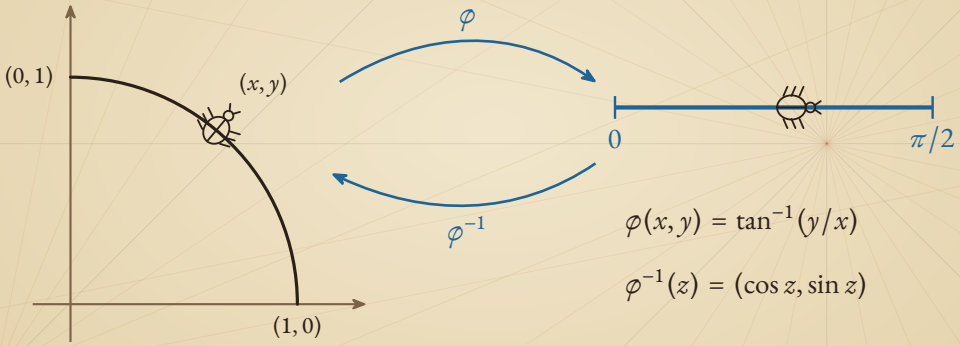




SECTION 3

Examples of manifolds

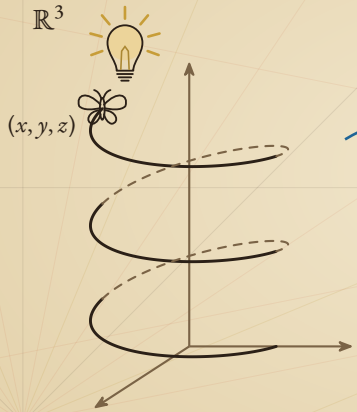
✳ *An arc of the circle*

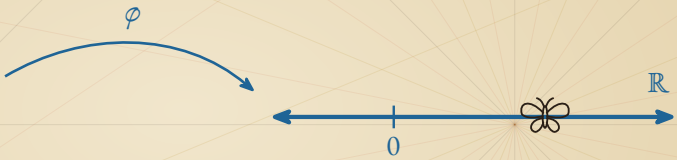


$$\varphi(x, y) = \tan^{-1}(y/x)$$

$$\varphi^{-1}(z) = (\cos z, \sin z)$$

✳ *A helix*

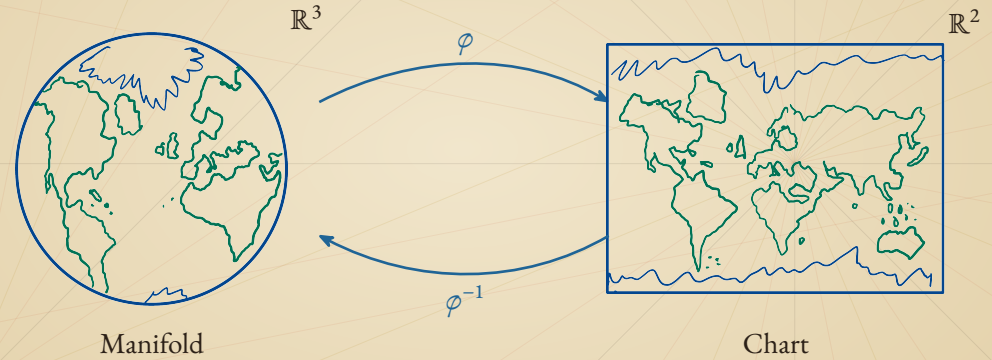




$$\varphi^{-1}(t) = (\cos t,\ \sin t,\ t)$$

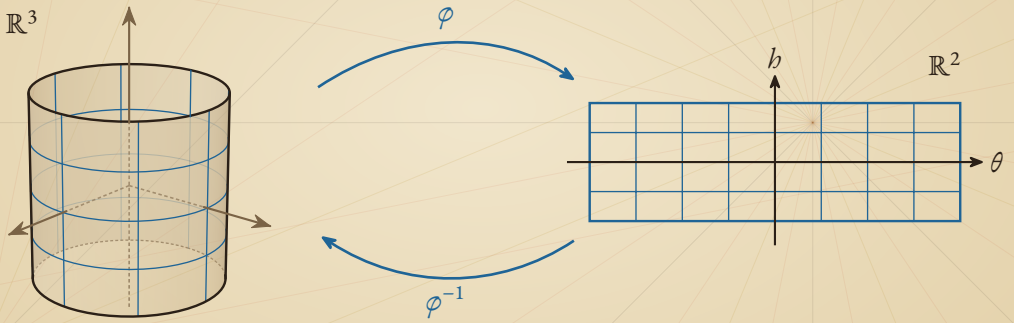
$$\varphi(x, y, z) = z$$

✳ *The globe and the world map*

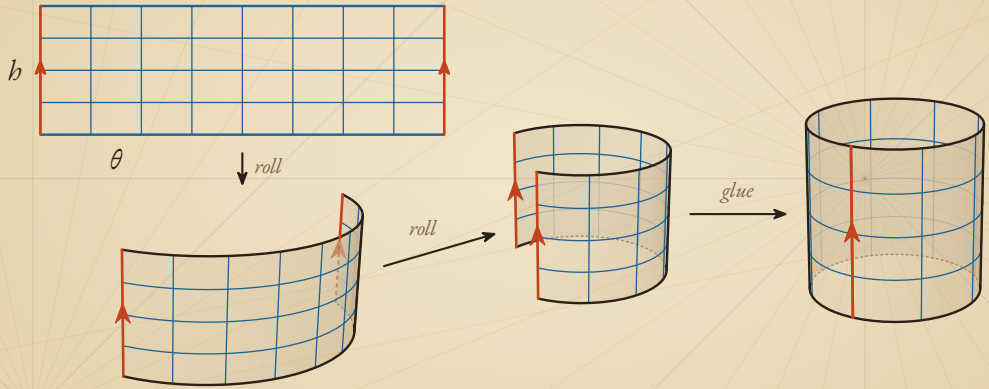


✳ *A cylinder*

$$\varphi^{-1}(\theta, h) = \left(\cos\tfrac{2\pi\theta}{360},\ \sin\tfrac{2\pi\theta}{360},\ h\right)$$



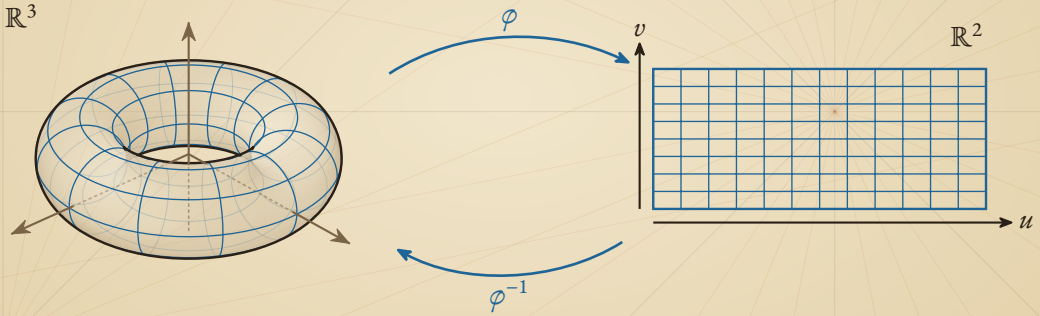
✳ *Folding a chart into a cylinder*



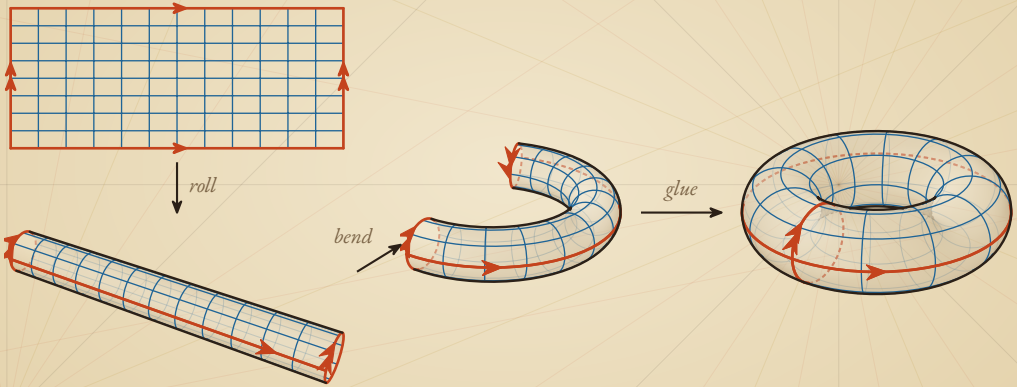
✳ *A torus*

$$\varphi^{-1}(u, v) = \big((R + r\cos v)\cos u,\ (R + r\cos v)\sin u,\ r\sin v\big)$$

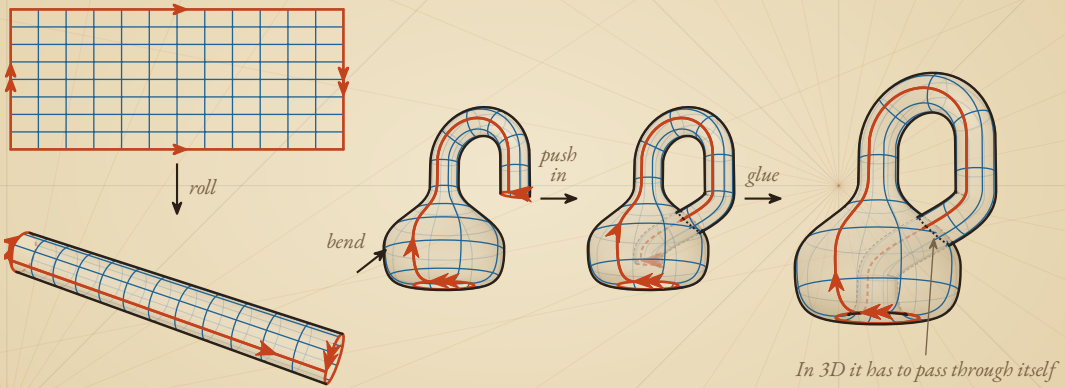
R: centre to the middle of the tube. r: radius of the tube.



✳ *Folding a chart into a torus*



✳ *Folding a chart into a Klein bottle*



✳ *Folding a chart into a projective plane*

